

# Hakan Tugrul Ota

PhD Student & Applied AI Researcher  
University at Albany SUNY, New York, United States  
hakan.otal@outlook.com — +1 (518) 608-9563 — linkedin.com/in/hakan-tugrul-otal  
hakanotal.github.io — github.com/hakanotal — huggingface.co/hotal

PhD Student and Research Assistant with a focus on quantifying and assessing risks of Generative AI systems. My research combines statistical modeling, ML, and cybersecurity to develop robust methodologies for AI risk evaluation and various applied AI projects. I bring hands-on experience in AI-driven risk engines for the insurance industry, published research on GenAI safety, and a strong foundation in experimental design and academic writing.

## INTERESTS

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AI Risk Assessment, GenAI Safety, Statistical Modeling, Cybersecurity, AI/ML, Large Language Models

## EDUCATION

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| <b>University at Albany, SUNY</b> , New York, United States<br>Ph.D. in Information Science<br>Data Analytics (Major) – Cybersecurity (Minor) | 08-2023 — Present<br><br>GPA: 3.9/4.0 |
| <b>University at Albany, SUNY</b> , New York, United States<br>M.S. in Information Science<br>Artificial Intelligence and Data Analytics      | 08-2023 — 05-2026<br><br>GPA: 3.9/4.0 |
| <b>Istanbul Technical University</b> , Istanbul, Turkey<br>B.S. in Computer Science                                                           | 09-2018 — 06-2023<br>GPA: 3.5/4.0     |

## EXPERIENCE

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| <b>University at Albany, SUNY</b><br><i>Graduate Research Assistant</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 08-2023 — Present |
| <ul style="list-style-type: none"><li>Conducted research on Gen-AI security and safety at TATAR Lab under the supervision of Dr. Unal Tatar.</li><li>Conducted research on integrating AI/LLM techniques for cyber threat detection at AI in Complex Systems Lab under the supervision of Dr. M. Abdullah Canbaz.</li><li>Built and evaluated LLM-based pipelines and model outputs in collaboration with faculty and lab members across interdisciplinary projects.</li><li>Supported various initiatives as a Graduate Assistant for Public Engagement and Enrollment.</li><li>Presented research at various conferences and workshops.</li><li>Teaching Assistant for CINF135: Concepts of AI.</li></ul> |                   |
| <b>CyRisk Inc.</b><br><i>AI/Software Engineer Intern</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 12-2025 — 05-2026 |
| <ul style="list-style-type: none"><li>Developed AI features for a cyber risk engine, including automated risk score calculation and analysis.</li><li>Worked on cybersecurity litigation cases leveraging AI-driven insights for the tool used by cyber insurers and underwriters.</li><li>Utilized API-driven development, deployment, and testing workflows with AWS, Docker, and Postman; coordinated with cross-functional teams on integration and issue resolution.</li></ul>                                                                                                                                                                                                                         |                   |
| <b>TATAR Lab (University at Albany, SUNY)</b><br><i>Summer Research Intern</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 05-2025 — 08-2025 |
| <ul style="list-style-type: none"><li>Worked on research projects related to AI security and safety.</li><li>Worked on developing 2025 Cyber Insurance Report for NAIC (National Association of Insurance Commissioners) by analyzing/charting annual data about policies and coverages.</li><li>Supervised student interns and led research group of undergraduate students.</li></ul>                                                                                                                                                                                                                                                                                                                     |                   |

**AI in Complex Systems Lab (University at Albany, SUNY)***Summer Research Intern*

05-2024 — 08-2024

- Worked on two research projects and published conference papers from them.
- Supervised undergraduate interns at UAlbany Summer Research Program (UASRP).

**Parieta NLP***Software Engineer*

01-2023 — 07-2023

- Designed and implemented frontend, backend, AI, and database for this legal tech startup.
- Developed the web application UI and integrated e-commerce features.
- Developed Nest.js backend services, MongoDB databases, and AI pipelines for the application.
- Deployed and maintained Docker containers on AWS infrastructure.

**Deka Technology***Software Engineer*

07-2022 — 12-2022

- Developed new API endpoints for enterprise-scale applications.
- Connected SAP services to backend systems through a customized integration layer.
- Worked with 100+ microservices using Jenkins, Openshift, Elastic, Redis, Quarkus, Apache Camel, and Keycloak.

**PoiLabs***Software Engineer Intern*

09-2021 — 10-2021

- Developed backend services for a mobile application for visually impaired people.
- Refactored the mobile application API with more dynamic and efficient approaches.
- Collaborated effectively using Bitbucket as VCS in a team setting.

**SKILLS**

- **Programming:** Python, C/C++
- **ML & Statistical Frameworks:** PyTorch, TensorFlow, Scikit-learn, NumPy, SciPy, Pandas
- **Data Visualization:** Matplotlib, Seaborn
- **Database Systems:** MongoDB, PostgreSQL, MySQL
- **Cloud:** Google Cloud (Cloud Storage, Compute Engine, Vertex AI), AWS (ECS, LightSail)
- **MLOps, CI/CD & Cloud:** Docker, Elasticsearch, Redis, AWS; Jenkins (CI/CD), Git VCS
- **LLM & GenAI Tools:** Hugging Face, Ollama, LMStudio, Unsloth, Llama-Factory, Vertex AI

**PUBLICATIONS**

- [1] M Abdullah Canbaz, Hakan Otal, Tugce Unlu, Nour Alhussein, and Brian Nussbaum. **Who Coordinates US Cyber Defense? A Co-Authorship Network Analysis of Joint Cybersecurity Advisories (2024–2025)**. *arXiv preprint arXiv:2510.20080*, 2025.
- [2] M Mikail Demir, Hakan T Otal, and M Abdullah Canbaz. **LegalGuardian: A Privacy-Preserving Framework for Secure Integration of Large Language Models in Legal Practice**. *arXiv preprint arXiv:2501.10915*, 2025.
- [3] Hao Jian Huang, Hakan T Otal, and M Abdullah Canbaz. **Federated learning in adversarial environments: Testbed design and poisoning resilience in cybersecurity**. In **2025 IEEE International Conference on Communications Workshops (ICC Workshops)**, pages 1079–1084. IEEE, 2025.
- [4] Hakan T Otal and M Abdullah Canbaz. **Ai-powered crisis response: Streamlining emergency management with llms**. In **2024 IEEE World Forum on Public Safety Technology (WFPST)**, pages 104–107. IEEE, 2024.
- [5] Hakan T Otal and M Abdullah Canbaz. **Llm honeypot: Leveraging large language models as advanced interactive honeypot systems**. In **2024 IEEE Conference on Communications and Network Security (CNS)**, pages 1–6. IEEE, 2024.

- [6] Hakan T Otal, Stephen V Faraone, and M Abdullah Canbaz. **A New Perspective on ADHD Research: Knowledge Graph Construction with LLMs and Network Based Insights.** In **International Conference on Complex Networks and Their Applications**, pages 337–349. Springer Nature Switzerland Cham, 2024.
- [7] Hakan T Otal, Eric Stern, and M Abdullah Canbaz. **Llm-assisted crisis management: Building advanced llm platforms for effective emergency response and public collaboration.** In **2024 IEEE Conference on Artificial Intelligence (CAI)**, pages 851–859. IEEE, 2024.
- [8] Hakan T Otal, Abdulhamit Subasi, Furkan Kurt, M Abdullah Canbaz, and Yasin Uzun. **Analysis of gene regulatory networks from gene expression using graph neural networks.** In **Digital Healthcare, Digital Transformation and Citizen Empowerment in Asia-Pacific and Europe for a Healthier Society**, pages 249–270. Academic Press, 2025.
- [9] Hakan T Otal, Elyse Zavar, Sherri B Binder, Alex Greer, and M Abdullah Canbaz. **Harnessing Deep Learning and Satellite Imagery for Post-Buyout Land Cover Mapping.** In **2024 IEEE Conference on Artificial Intelligence (CAI)**, pages 665–671. IEEE, 2024.

## SELECTED PROJECTS

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### Autonomous Cyber-Offensive LLM Agents (VICEROY Project) 2026

- Leveraged Vertex AI to deploy large-scale, autonomous agents using a hybrid of frontier models and fine-tuned cybersecurity open-source models on Google Cloud infrastructure.
- Advanced the D-CIPHER multi-agent framework and contributed to the autonomous Cyber-Offensive LLM Agent research and red-teaming capability assessments of agents.
- Authored a research paper for presentation to DoW cyber spectrum leaders at the VICEROY Symposium.

### NAIC’s 2025 Cyber Insurance Market Report 2025

- Contributed to the annual cyber insurance market analysis for the National Association of Insurance Commissioners (NAIC) by integrating and reconciling data from the Cybersecurity and Identity Theft Supplement and the International Insurers Department (IID).
- Synthesized multi-source regulatory and market data into statistical analyses and visual narratives on trends, policy coverages, and loss ratios for policy and industry audiences.

### ATTASH - Anticipating Terrorist Technology Adoption to Secure the Homeland 2024

- Contributed to the ATTASH Tool Pilot as part of the development team.
- Developed Monte Carlo simulations in Excel and supporting Python scripts for statistical analysis.

### Distributed Network Load Generation Software 2023

- Bachelors Graduation Project

## SERVICES

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### IEEE

IEEE Student Member 09-2023 — Present

IEEE Computational Intelligence Society (CIS) Member 03-2024 — Present

### ACM

Student Member 10-2024 — Present

### University at Albany, SUNY

Member of AI in Complex Systems Lab 08-2023 — Present

Member of TATAR Lab 05-2025 — Present

Graduate Assistant of Assistant Dean for Public Engagement and Enrollment Spring 2025

Supervisor of Student Interns and VICEROY Research Group at TATAR Lab Summer 2025

Supervisor of Undergrad. Interns at UAlbany Summer Research Program (UASRP) Summer 2024

Teaching Assistant for ‘CINF135: Concepts of AI’ Spring 2024, Fall 2024

Vice President, UAlbany Turkish Student Association 03-2026 — Present

## AWARDS & GRANTS

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### University at Albany, SUNY

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|-----------------------------------------------------------------------------------------------------------------------------|---------------------|
| The Catalyst Award for Faculty Mentorship of Undergraduate Research,<br>Scholarly Inquiry, and Creative Endeavors (Nominee) | 2026                |
| Student Travel Grant to attend RSAC 2026                                                                                    | (\$900) — 2026      |
| Ph.D. Full Tuition Scholarship and Stipend                                                                                  | (\$32K) — 2023-2028 |

### SUNY AI Platform – Google Cloud

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| Cloud computing grant via SUNY AI Platform (Google Cloud) | (\$30K) — 2026 |
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### RISE Fellowship

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| Fellowship at Research and Innovators Startup Exchange (RISE) entrepreneurship program | (\$1290) — 2025 |
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### IEEE Computational Intelligence Society

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| Student Travel Grant for IEEE CAI | (\$1200) — 2024 |
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### IEEE Communications Society

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| Student Travel Grant for IEEE CNS | (\$800) — 2023 |
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## WORKSHOPS & TALKS

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|---------------------------------------------------------------|------|
| Hot Topics in the Science of Security Symposium (HotSoS) Talk | 2026 |
| NTIR Workshop — “Building a Chatbot with RAG from Scratch”    | 2026 |
| ACM UAlbany Workshop — “Deploying Models on the Edge”         | 2025 |

## REVIEWED (TPC)

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|--------------------------------------------------------------------|------------|
| Conference on Computational Linguistics (COLING)                   | 2025       |
| Information Security Journal: A Global Perspective                 | 2024       |
| IEEE Conference on Communications and Network Security (CNS)       | 2024       |
| IEEE Conference on Local Computer Networks (LCN)                   | 2024       |
| IEEE International Conference on Computer Communications (INFOCOM) | 2024       |
| Complex Networks and their Applications (CNA)                      | 2023, 2024 |

## Recommendation Letters Available upon Request:

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### Dr. M Abdullah Canbaz

*Assistant Professor*

*Department of Information Sciences and Technology*

*University at Albany SUNY, NY, USA*

E-mail: mcanbaz@albany.edu

Website: albany.edu/cehc/faculty/m-abdullah-canbaz

### Dr. Unal Tatar

*Associate Professor*

*Department of Cybersecurity*

*University at Albany SUNY, NY, USA*

E-mail: utatar@albany.edu

Website: albany.edu/cehc/faculty/unal-tatar